كلية هندسة الذكاء الاصطناعي
FACULTY OF ARTIFICIAL INTELLIGENCE ENGINEERING
الجامعة السورية الخاصة
SYRIAN PRIVATE UNIVERSITY

AURA HEALTH HUB: A Multi-Tenant EHR System with AI-Powered Audio Transcription & ETL Data Integration

student:

Maiss Mansour
Amal Abo Althahab

Supervised by:
Dr. Mouhib Alnoukari
Eng. Anas Abdulaziz

:Damascus

Table of Contents

Table of Contents ..... **II**
List of Tables ..... **III**
List of Figures ..... **IV**
Chapter 1: Introduction **1**
1-1- Introduction 1
1-1-1- System Definition 1
1-2- Problem Statement and Scientific Problem 1
1-2-1- Problem Statement 1
1-2-2- Scientific Problem 2
1-3- Project Objectives 2
1-4- Contribution of the Project 2
1-5- Beneficiaries of the Project 2
1-6- Organization of the Thesis 2
Chapter 2: Theoretical Study **3**
2-1- Electronic Health Records (EHR) 3
2-2- Multi-Tenant Architecture 4
2-3- ETL and Data Integration 4
2-4- Automatic Speech Recognition (ASR) 5
2-5- Speaker Diarization 5
2-6- Clinical Note Generation and Summarization 6
2-7- Security, Privacy, and Access Control 6
Chapter 3: Reference Study **9**
3-1- Reference Systems 9
3-2- Modern Techniques and Frameworks in the Field 9
3-3- Comparative Analysis of Systems 9
3-3-1- Comparison Method 9
3-3-2- Feature Comparison Matrix 10
3-3-3- Analysis 10
3-4- Research Gap and Proposed Innovation 10
Chapter 4: Research Methodology (Solution Idea) **11**
4-1- Proposed Solution 11
4-2- Box Diagram 11
4-3- Flowchart: AI Audio Processing Engine 12
4-4- Flowchart: Patient Record Integration 13

4-5- Flowchart: ETL Data Integration 14

List of Tables

Table 3.1: Feature comparison between AURA HEALTH HUB and the reference systems 10

List of Figures

Figure 4.1: AURA HEALTH HUB — Box Diagram 12
Figure 4.2: AI Audio Processing Engine 13
Figure 4.3: Patient Record Integration 14
Figure 4.4: ETL Data Integration 15

Chapter 1: Introduction

1-1- Introduction

AURA HEALTH HUB is a unified medical record platform that brings together patient data from healthcare providers across the Syrian Arab Republic.

1-1-1- System Definition

The proposed system is a cloud-based Unified Electronic Health Record (EHR) platform designed to centralize patient medical histories across hospitals, clinics, and pharmacies. To enhance security and workflow efficiency, it incorporates:

- a **Multi-Tenant Architecture** for data isolation,
- an automated **ETL pipeline** for data ingestion,
- **AI-driven audio processing** for clinical documentation.

In the future, and subject to the required specifications, regulations, and safeguards being in place, the project will be capable of making this medical data available for scientific research purposes.

1-2- Problem Statement and Scientific Problem

1-2-1- Problem Statement

Patient medical records are scattered across independent healthcare providers (hospitals, clinics, and pharmacies). Each provider stores data in its own system and format, ranging from legacy SQL databases to Excel sheets, Word documents, PDFs, and CSV files. As a result:

- **No unified patient history:** a doctor treating a patient cannot access the diagnoses, test results, and prescriptions recorded elsewhere, so medical history is lost or incomplete.
- **Repeated, unnecessary diagnostic tests:** tests already performed at another provider are repeated, increasing cost, delay, and patient burden.

- **Heterogeneous and unstructured data:** records in different formats and schemas cannot be easily merged or queried, and manual re-entry is slow and error-prone.
- **Heavy documentation workload:** doctors spend a significant part of each consultation typing notes manually, which reduces time with the patient and lowers note quality and consistency.
- **Privacy and security risk:** centralizing sensitive data from many institutions requires strict isolation between organizations and full traceability of every access.

1-2-2- Scientific Problem

How to build a secure, centralized EHR that can (a) ingest and normalize heterogeneous, unstructured medical data from multiple sources, (b) keep each institution's data isolated and private, and (c) automate clinical documentation from doctor–patient conversations using speech recognition and speaker separation.

1-3- Project Objectives

- Centralize patient history into a single, accessible EHR.
- Guarantee data isolation and privacy for hospitals using multi-tenancy.
- Automate data ingestion from unstructured files (Word/Excel) using an ETL pipeline.
- Streamline clinical documentation via speaker diarization and live audio processing.
- Reduce duplicate diagnostic tests by giving authorized providers access to the full patient history.
- Enforce role-based access control and audit logging for every access to patient data.
- Prepare the platform to support scientific research in the future, once the required specifications, regulations, and safeguards are in place.

1-4- Contribution of the Project

1-5- Beneficiaries of the Project

1-6- Organization of the Thesis

Chapter 2: Theoretical Study

This chapter presents the theoretical concepts on which AURA HEALTH HUB is built and that are needed during its implementation: electronic health records and medical data standards, multi-tenant architecture, ETL data integration, automatic speech recognition, speaker diarization, clinical text generation and summarization, and the security and privacy foundations of a health information system.

2-1- Electronic Health Records (EHR)

An Electronic Health Record is a digital version of a patient's medical history. It typically contains demographics, diagnoses, medications, allergies, laboratory and imaging results, visit notes, and treatment plans. Unlike a paper file or a record kept inside one institution's isolated system, a unified EHR lets every authorized provider see the same up-to-date history of the patient.

The main benefits of a unified EHR are:

- A complete and consistent medical history available at the point of care.
- Fewer duplicated tests and fewer medication errors, because previous results and prescriptions are visible.
- Faster access to information and better coordination between hospitals, clinics, and pharmacies.
- A structured data foundation that can later support analytics and scientific research.

Medical data standards. To exchange and compare data between systems, health information systems rely on common standards. The most relevant ones for this project are: HL7 FHIR (a standard for representing and exchanging health data as resources through web APIs), ICD-10 (a classification of diseases and diagnoses), SNOMED CT (a clinical terminology), and LOINC (codes for laboratory and clinical observations). Mapping local data to such standards is what allows records from different providers to be merged meaningfully.

2-2- Multi-Tenant Architecture

Multi-tenancy is an architecture in which a single instance of the software serves multiple customers (tenants), here the hospitals, clinics, and pharmacies, while keeping each tenant's data logically isolated from the student. It reduces cost and maintenance compared to deploying a separate system for every institution.

There are three common data-isolation models:

- **Separate database per tenant:** the strongest isolation, but the highest cost and management effort.
- **Shared database, separate schema per tenant:** a balance between isolation and cost.
- **Shared database and shared schema:** all tenants share the same tables and each row carries a tenant identifier. It is the cheapest and most scalable model, and requires strict enforcement of tenant filtering, for example through row-level security.

The choice depends on the required isolation level, the number of tenants, and the operational cost. In every model, each request must be tied to an authenticated tenant context so that one institution can never read another institution's data without authorization.

2-3- ETL and Data Integration

ETL (Extract, Transform, Load) is the process of collecting data from heterogeneous sources and bringing it into a target system. In AURA HEALTH HUB it is used to ingest legacy data into the central database:

- **Extract:** read data from legacy SQL databases, Excel and CSV files, Word documents, and PDFs. Semi-structured and unstructured files require parsing, and scanned documents may require optical character recognition (OCR).
- **Transform:** clean the data, unify formats (dates, units, names), remove duplicates, and map each source schema to the standard target schema, including medical coding where possible.

- **Validate:** check data quality (required fields, valid ranges and types, referential integrity). Invalid records are logged and sent for retry or manual correction instead of being loaded.
- **Load:** insert the validated records into the central database, with each record assigned to the correct tenant.

A further challenge is patient identity matching: the same patient may appear with different spellings or incomplete identifiers across sources, so records must be matched carefully to avoid wrongly merging two patients or splitting one patient into two.

2-4- Automatic Speech Recognition (ASR)

Automatic Speech Recognition converts spoken audio into text. A classical ASR system consists of an acoustic model (maps audio features to phonetic units), a pronunciation lexicon, and a language model (scores likely word sequences). Modern systems are mostly end-to-end deep neural networks, based on architectures such as Transformers, that map audio directly to text and are trained on large amounts of speech data.

Challenges relevant to this project include background noise in clinics, overlapping speech, medical terminology, and the variety of Arabic dialects and code-switching between Arabic and English medical terms. These affect the choice of model and the need for domain adaptation and for human review of the result.

2-5- Speaker Diarization

Speaker diarization answers the question “who spoke when?” in a recording with more than one speaker. In a consultation it separates the doctor’s speech from the patient’s, so the transcript can be attributed correctly (for example, the symptoms come from the patient and the diagnosis from the doctor). A typical pipeline has the following stages:

- Voice activity detection to find the segments that contain speech.
- Segmentation of the audio into short speaker-homogeneous parts.

- Extraction of a speaker embedding (a numerical voice fingerprint) for each segment, using neural embeddings such as x-vectors.
- Clustering of the embeddings so that each cluster corresponds to one speaker.

The standard evaluation metric is the Diarization Error Rate (DER), which combines missed speech, false alarms, and speaker confusion. In this project the diarization output is combined with the ASR output to produce a transcript labelled by speaker.

2-6- Clinical Note Generation and Summarization

After transcription and speaker separation, the text must be turned into a structured clinical note. This uses Natural Language Processing (NLP): information extraction identifies medical entities such as symptoms, diagnoses, medications, and follow-up steps, and text summarization condenses the consultation into its key findings. Large Language Models (LLMs) are well suited to these tasks, and the output can follow a clinical note structure such as SOAP (Subjective, Objective, Assessment, Plan).

Because generated text can contain errors or omissions, and the consequences in medicine are serious, the design includes a human-in-the-loop step: the doctor reviews, corrects, and approves the note before it is stored in the patient's record.

2-7- Security, Privacy, and Access Control

Medical data is among the most sensitive personal data, so security is a core requirement and not an optional feature. The main mechanisms are:

- **Encryption:** data is encrypted at rest in the database and in transit between clients and servers (TLS).
- **Authentication and authorization:** users must prove their identity, and Role-Based Access Control (RBAC) limits what each role (doctor, patient, pharmacist, administrator) may see or do.
- **Audit logging:** every access to or change of patient data is recorded with who, what, and when, so misuse can be detected and investigated.

- **Tenant isolation:** as described in Section 2-2, preventing one institution from accessing another's data.
- **Data minimization and consent:** only the data needed for care is shown, and any future use of the data for research requires anonymization and the legal and ethical approvals in force.

International frameworks such as HIPAA and GDPR provide useful reference principles for protecting health data. The system must also follow the laws and regulations that apply in the Syrian Arab Republic.

Chapter 3: Reference Study

This chapter compares AURA HEALTH HUB with existing electronic health record systems, national health platforms, and AI documentation products, in order to show which capabilities already exist and which gap the project addresses.

3-1- Reference Systems

- **OpenMRS:** an open-source EHR platform widely used in resource-limited healthcare settings.
- **OpenEMR:** open-source EHR and practice-management software, mainly for clinics.
- **Epic / Oracle:** commercial enterprise EHR systems used by large hospitals and hospital networks.
- **Estonia e-Health:** a national health information system that gives providers access to a nationwide patient record.
- **NABIDH / Sehhaty:** regional national health platforms: NABIDH is Dubai's health information exchange, and Sehhaty is Saudi Arabia's national health application.
- **DAX / Abridge:** ambient AI products that capture the consultation and draft clinical notes for the clinician.

3-2- Modern Techniques and Frameworks in the Field

3-3- Comparative Analysis of Systems

3-3-1- Comparison Method

The matrix in Section 3-3-2 is taken from the comparison prepared in the project user-story file. The symbols keep their original meaning: ✓ supported, ◐ partial or conditional support, and ✗ not supported in that matrix. It is a planning comparison made by the project team, not an independent audit of the current versions of these products.

3-3-2- Feature Comparison Matrix

Table 3.1: Feature comparison between AURA HEALTH HUB and the reference systems

Feature	OpenMRS	OpenEMR	Epic / Oracle	Estonia e-Health	NABIDH / Sehhaty	DAX / Abridge	AURA
New patient registration	✓	✓	✓	◐	✓	✗	✓
Unified cross-provider record	◐	✗	◐	✓	✓	✗	✓
Multi-tenant data isolation	✗	✗	◐	◐	◐	◐	✓
Role-based access control	✓	✓	✓	✓	✓	✓	✓
Clinical encounter documentation	✓	✓	✓	◐	◐	◐	✓
Allergies and chronic conditions	✓	✓	✓	✓	✓	✗	✓
Laboratory and results management	✓	✓	✓	◐	✓	✗	✓
Electronic prescribing	◐	✓	✓	✗	✓	✗	✓
Pharmacy dispensing tracking	◐	◐	✓	✗	✓	✗	✓
Legacy Word/Excel migration	◐	◐	◐	✗	✗	✗	✓
Import validation, review and rollback	✗	✗	◐	✗	✗	✗	✓
Patient portal	◐	✓	✓	✓	✓	✗	✓
Patient can see who accessed record	✗	✗	◐	◐	✓	✗	✓
Immutable audit trail	✓	✓	✓	✓	✓	✓	✓
Speech-to-text transcription	✗	✗	◐	✗	✗	✓	✓
Speaker diarization	✗	✗	◐	✗	✗	✓	✓
AI clinical-session summary	✗	✗	◐	✗	✗	✓	✓
Mandatory clinician review of AI output	✗	✗	◐	✗	✗	✓	✓
Arabic and RTL support	◐	◐	◐	✗	✗	✗	✓

3-3-3- Analysis

In the supplied matrix, core EHR capabilities appear across several systems, while the emphasis differs: open-source and commercial EHRs focus on clinical records, national exchanges focus on cross-provider sharing, and ambient documentation products focus on consultation capture and summarization. AURA aims to combine unified records, tenant isolation, legacy migration, Arabic audio documentation, and clinician review in one platform. A check mark is not proof of quality, security, or regulatory compliance; these require separate verification.

3-4- Research Gap and Proposed Innovation

Chapter 4: Research Methodology (Solution Idea)

4-1- Proposed Solution

AURA HEALTH HUB is a centralized, cloud-based, multi-tenant Electronic Health Record platform that integrates patient data across hospitals, clinics, and pharmacies and automates clinical documentation using AI. The solution has three core components:

- **ETL pipeline:** extracts data from multiple sources, transforms it to a standard schema, validates it, and loads it into the central database.
- **Multi-tenant encrypted central database:** stores patient records, medical history, and audit logs with isolation between tenants and role-based access (doctors, patients, pharmacists, administrators).
- **AI audio engine:** speech-to-text, speaker diarization, structured clinical note generation, and auto-summarization, with doctor review before storage.

4-2- Box Diagram

The diagram shows the main building blocks of the system and how data flows between them: data sources feed the ETL pipeline, the AI audio engine and the ETL pipeline both write to the central multi-tenant encrypted database, and the database serves the different user roles.

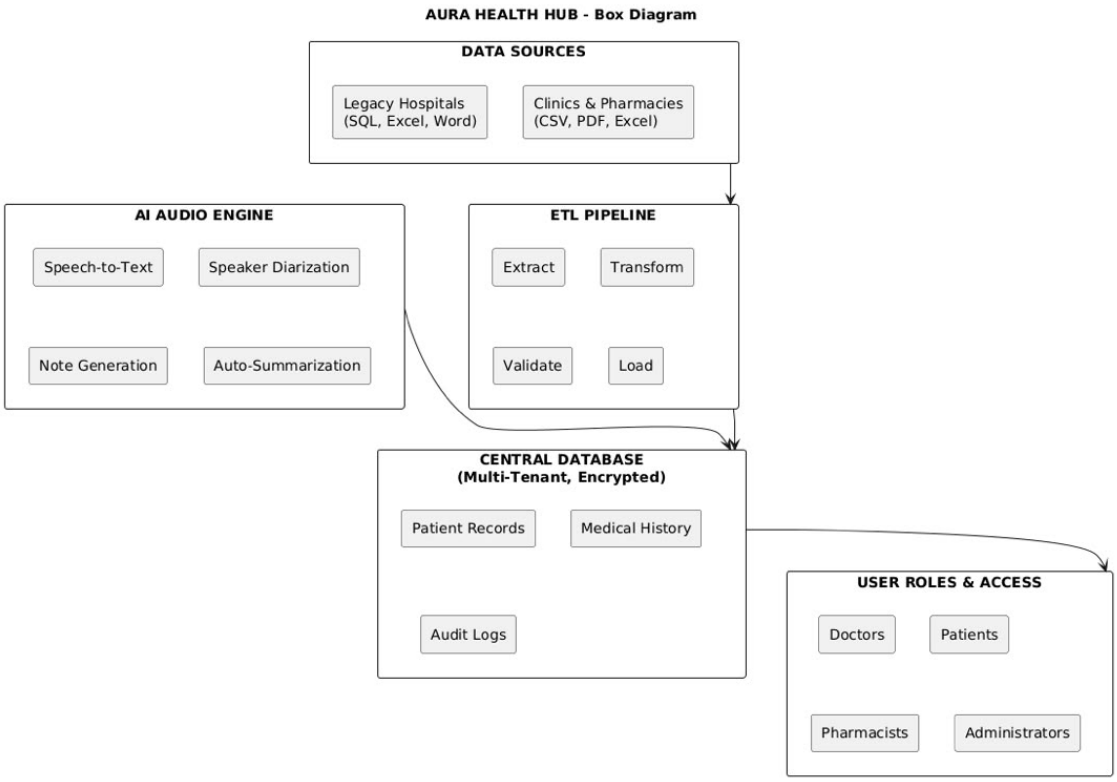


Figure 4.1: AURA HEALTH HUB — Box Diagram

4-3- Flowchart: AI Audio Processing Engine

The consultation audio is sent to the AI engine, where speech-to-text and speaker diarization run in parallel. Their results are combined into structured clinical notes and a summary, which the doctor must verify and approve before the note is stored in the patient record.

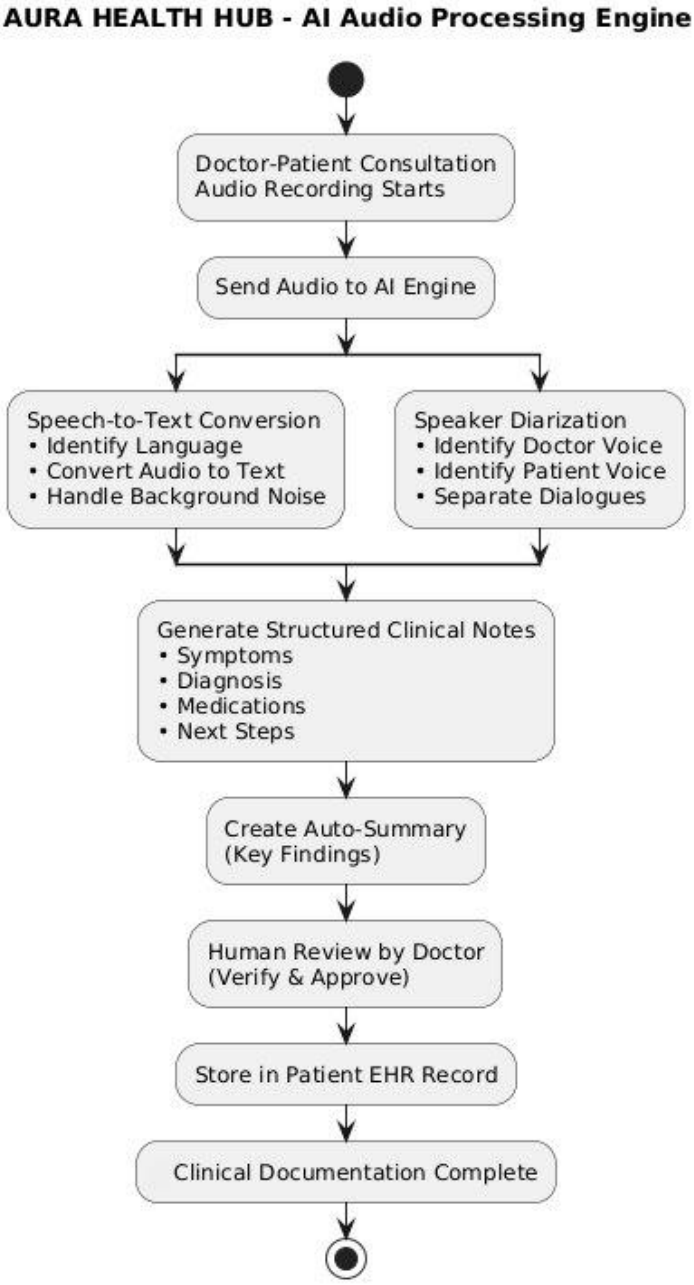


Figure 4.2: AI Audio Processing Engine

4-4- Flowchart: Patient Record Integration

The visit is documented either by recording the live consultation or by manual entry by the doctor. The data goes through the AI audio engine, is stored in the isolated multi-tenant database, and becomes available to authorized providers, which avoids duplicate tests.

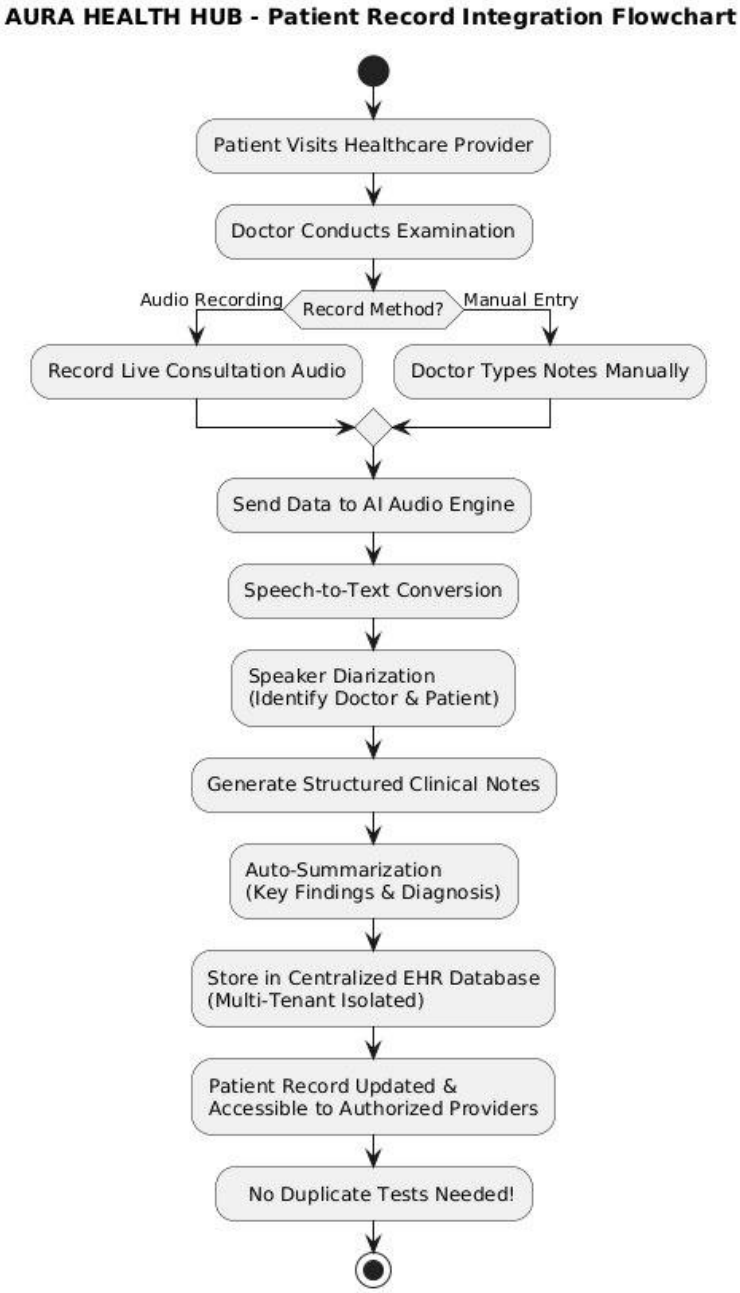


Figure 4.3: Patient Record Integration

4-5- Flowchart: ETL Data Integration

Legacy data from several sources is extracted, transformed to the standard schema, and validated. Valid records are loaded into the central database; invalid records are logged, the administrator is alerted, and the record is retried or corrected manually.

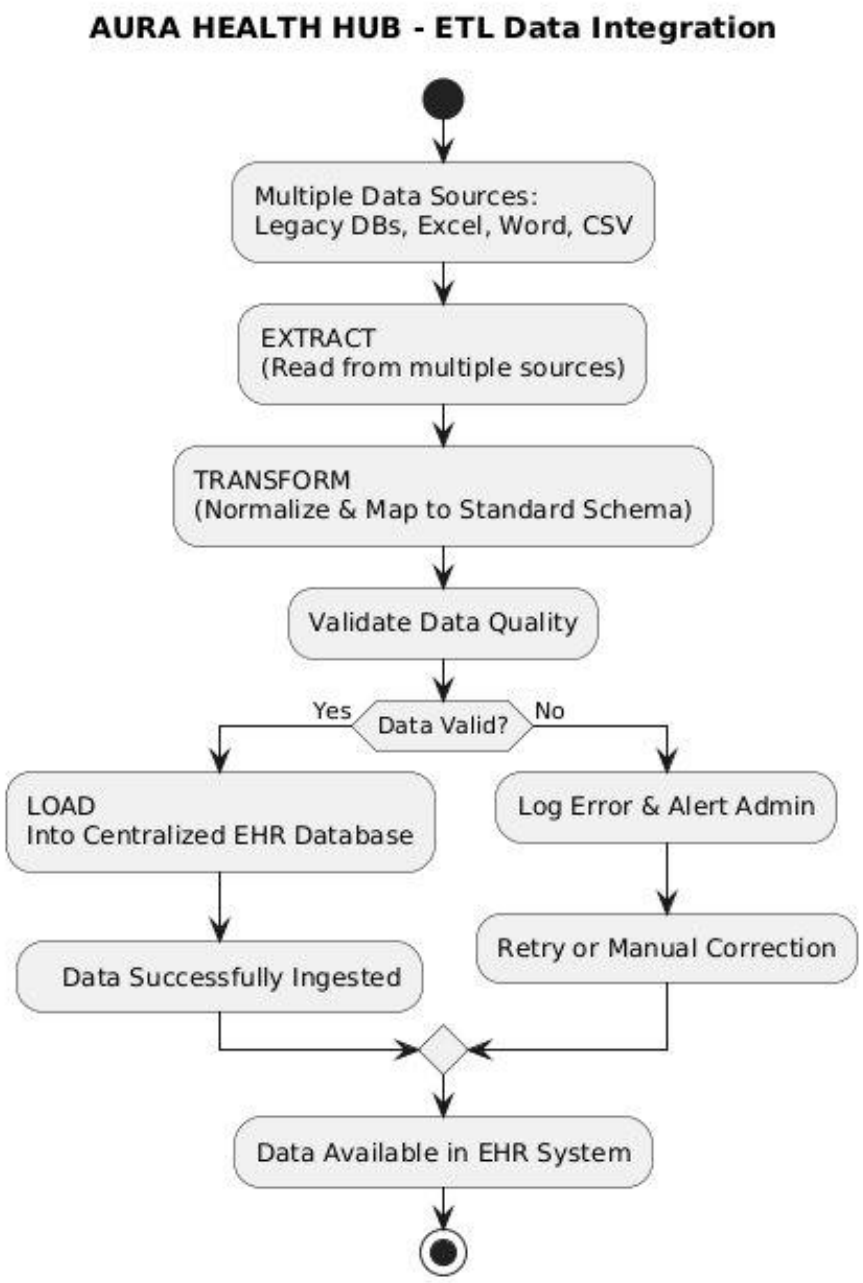


Figure 4.4: ETL Data Integration